DEPARTMENT OF COMPUTER ENGINEERING
Course: CPE342 Operating Systems
Deadline: November 02, 2026, 23:59

Mini-Project: CPU Scheduling Algorithms Simulation (FCFS, SJF, RR)

1. Assignment Description & Objective

Build a discrete-event CPU scheduling simulator comparing First-Come First-Served (FCFS), Shortest Job First (SJF), and Round Robin (RR) algorithms.

The simulator should read process arrival times and burst times from a CSV file, compute average waiting time, turnaround time, and output a visual ASCII Gantt chart.

Analyze the impact of different time quantum values on Round Robin context switching overhead.

2. Grading & Evaluation Criteria

• Algorithm implementation correctness (45%)

• Simulation metrics computation & Gantt chart output (30%)

• Comparative experimental report (25%)

Note: Please deconstruct this assignment into daily actionable tasks using TaskFlow AI to avoid procrastination and burnout.